Final Project

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```
# install.packages("tidyverse")   # run once in the console if tidyverse is not installe
d; left commented so the notebook knits
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.2.1     ✔ readr     2.2.0
## ✔ forcats   1.0.1     ✔ stringr   1.6.0
## ✔ ggplot2   4.0.3     ✔ tibble    3.3.1
## ✔ lubridate 1.9.5     ✔ tidyr     1.3.2
## ✔ purrr     1.2.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
to become errors
```

```
library(ggplot2)
```

Introduction

Hi everyone! This is the progstats final project of Kirya, Chris, and Jerry

Kirya: PREP Scholar in Harwell lab, studying developmental neurobiology. I am interested in the evolution and neural mechanism of behavior! Chris: PROPEL Scholar in Shoichet lab, studying molecular docking for drug discovery. I get excited by anything biophysical chemistry related, including protein interactions! Jerry: PROPEL scholar in DelRosso lab, studying gene regulation. I am broadly interested in transcription, genome organization, and intrinsically disordered proteins!

Maximum lifespan varies enormously across mammals, from shrews that live only a few years to elephants and primates that live for decades. In general, larger mammals tend to live longer, but some clades, like primates and bats, live longer than expected for their body size. This raises a larger question: what drives differences in longevity across clades — shared evolutionary history, or ecological and morphological factors like body size?

maxiumum lifespan varies enormously across mammals, from shrews that live only a few years to elephants that live for decades. The general wisdom is that larger mammals tend to live longer, but certain clades like priamtes and bats live longer than expected for their body size. This raises a larger question: what dirves differences in longevity across clades… shared evolutionary history, or ecological and morpholigical factors like body size? This sort of investigation may help us to understand the origin of longevity itself, and provide insights into how and why some humans live longer, healthier, than others.

To investigate this, we combined two open-source datasets: PanTHERIA, which has morphological, life history, and ecological traits for thousands of mammal species, and AnAge, which has markers of aging across species, including maximum longevity. I use taxonomic Order as a rough measure of shared evolutionary history and adult

body length as the main morphological factor.

Research question: Across mammals, does adult body length predict maximum longevity, and do primates and carnivores differ in maximum longevity, even when body length is taken into account?

Dataset background and limitations

PanTHERIA is a dataset that was compiled from previously published museum records, literature, and databases over a span of three years. As of 2009, it inclued nearly all known living and recently extinct animal species. Anage is also a compiled dataset that relies on previously published literature. It is constantly being updated and currently has over 4000 species.

PanTHERIA includes many different variables, including juvenile and adult morphological variables, gestation length, weaning age, litter size, diet, geography, climate, and more. Anage contains data that is specifically geared towards aging research; longevity, male and female age at sexual maturity, metabolic rate, etc.

Due to the fact that these datasets are assembled from preexisting literature, they are vast, but come with many limitations. One of these limitations is most species are not represented across all measures and variables. for example, many species do not have data for female age at sexual maturity, group composition and size, habitat layer, etc. We dealt with this by limiting our analysis to only species that were represented by all variables of interest to us, which likely produced a bias in our data towards only species that are already well-studied. This therefore may not be representative of the entire animal kingdom. Another limitation associated with these datasets is that they do not account for differences in methods for collecting the raw data. This may vary study-to-study, and differences in methods are often not noted/recorded. This is highlighted by the sample size and specimen origin columns of the data table, which demonstrate that some data comes from captive populations while others wild, and some species have larger sample sizes than others. This almost surely biases that data, but we did not perform any measures to control for this here due to the preliminary nature of these analyses.

Original sources:

- PanTHERIA: Jones, K. E., et al. (2009). PanTHERIA: a species-level database of life history, ecology, and geography of extant and recently extinct mammals. *Ecology*, 90(9), 2648.
- AnAge: Tacutu, R., et al. (2018). Human Ageing Genomic Resources: new and updated databases. *Nucleic Acids Research*, 46(D1), D1083–D1090. https://genomics.senescence.info/species/ (https://genomics.senescence.info/species/)

Part 1:

Filter and select data

in the code below, I am writing a csv from the PanTheria open-source dataset, which has morphological and life history parameters for thousands of species. I then subset it. relevant columns: Genus, species, adult body mass, adult body length including head, gestation length, and midrange latitude. Adult body length and body mass are

morphological factors, gestation length is a life history factor, and midrange latitude is an ecological factor. Removing any rows with “NA” in any of these columns to get a subset of the data that has no missing values and is workable.

```
url <- "https://raw.githubusercontent.com/pedroj/Megafauna/master/datasets/PanTHERIA_1-0_WR05_Aug2008.txt"
pantheria_data <- read.delim(url, header = TRUE, sep = "\t", na.strings = "-999")
write.csv(pantheria_data, "PanTheria_WR05_mammals.csv", row.names = FALSE)
print("csv file has been created in working directory")
```

```
## [1] "csv file has been created in working directory"
```

```
# View(pantheria_data)
cleaned_pantheria <-
  pantheria_data %>%
  select(MSW05_Genus, MSW05_Species, X5.1_AdultBodyMass_g, X13.1_AdultHeadBodyLen_mm, X9.1_GestationLen_d, X26.4_GR_MidRangeLat_dd) %>%
  filter(X5.1_AdultBodyMass_g != "NA", X13.1_AdultHeadBodyLen_mm != "NA", X9.1_GestationLen_d != "NA", X26.4_GR_MidRangeLat_dd != "NA")
# View(cleaned_pantheria)
```

Joins

using AnAge database, which examines different markers of animal aging across species, including maximum longevity, my outcome variable, and taxonomic Order, which I use as a rough measure of shared evolutionary history. in the code blocks below, I first load in the anage csv. I then make sure that the columns I want to join by, species and genus, agree with each other across datasets by renaming the columns of cleaned_pantheria MSW05_Genus and MSW05_Species to Genus and Species and putting this into a new df, renamed_pantheria. Finally, I join anage and renamed_pantheria under agingpantheria, and move the taxonomy columns to the front.

```
anage_url <- "https://gist.githubusercontent.com/Audree/4ef1ebd18ed0d9e9f5a5/raw/"
anage <- read_csv(anage_url)
```

```
## Rows: 968 Columns: 16
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## chr (11): Kingdom, Phylum, Class, Order, Family, Genus, Species, Common name...
## dbl  (5): HAGRID, Female maturity (days), Male maturity (days), Gestation/In...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# View(anage)
```

```
renamed_pantheria <- cleaned_pantheria %>%
rename(Genus = MSW05_Genus)%>%
rename(Species = MSW05_Species)

agingpantheria <- renamed_pantheria %>%
  inner_join(anage, by = c("Genus", "Species"))

# move the taxonomy columns to the front, from broadest to most specific
agingpantheria <- agingpantheria %>% relocate(Kingdom, Phylum, Class, Order, Family, Gen
us, Species)

# View(agingpantheria)
glimpse(agingpantheria)
```

```
## Rows: 289
## Columns: 20
## $ Kingdom                      <chr> "Animalia", "Animalia", "Animalia", "Ani…
## $ Phylum                       <chr> "Chordata", "Chordata", "Chordata", "Cho…
## $ Class                        <chr> "Mammalia", "Mammalia", "Mammalia", "Mam…
## $ Order                        <chr> "Carnivora", "Carnivora", "Carnivora", "…
## $ Family                       <chr> "Canidae", "Canidae", "Canidae", "Canida…
## $ Genus                        <chr> "Canis", "Canis", "Canis", "Canis", "Bos…
## $ Species                      <chr> "adustus", "aureus", "latrans", "lupus",…
## $ X5.1_AdultBodyMass_g         <dbl> 10392.49, 9658.70, 11989.10, 31756.51, 8…
## $ X13.1_AdultHeadBodyLen_mm    <dbl> 745.32, 827.53, 872.39, 1055.00, 2700.00…
## $ X9.1_GestationLen_d          <dbl> 65.00, 61.24, 61.74, 63.50, 273.75, 62.5…
## $ X26.4_GR_MidRangeLat_dd      <dbl> -6.00, 21.14, 39.70, 47.38, 15.43, -9.18…
## $ HAGRID                       <dbl> 2522, 2523, 2525, 2526, 2358, 2527, 3126…
## $ `Common name`                <chr> "Side-striped jackal", "Golden jackal", …
## $ `Female maturity (days)`     <dbl> 274, 334, 274, 669, 590, 309, 1461, 1515…
## $ `Male maturity (days)`       <dbl> 274, 334, 274, 669, 550, 251, 1826, 1826…
## $ `Gestation/Incubation (days)` <dbl> 64, 63, 63, 62, 275, 60, 184, 227, 232, …
## $ `Maximum longevity (yrs)`    <dbl> 13.7, 18.8, 21.8, 20.6, 26.2, 16.7, 37.0…
## $ `Specimen origin`            <chr> "captivity", "captivity", "captivity", "…
## $ `Sample size`                <chr> "small", "medium", "medium", "large", "m…
## $ `Data quality`               <chr> "acceptable", "acceptable", "acceptable"…
```

grouping and summaries

we decided to group by taxonomic Order, to see how shared evolutionary history might affect maximum longevity and body length. I also count the number of species in each order and sort from most to least species, to see which orders are well-represented enough to compare. I am putting this into a new df called order_summaries.

```
order_summaries <-
agingpantheria %>%
  group_by(Order) %>%
  summarize(
    n_species = n(),
    median_max_longevity = median(`Maximum longevity (yrs)`, na.rm = TRUE),
    mean_max_longevity = mean(`Maximum longevity (yrs)`, na.rm = TRUE),
    median_body_length = median(X13.1_AdultHeadBodyLen_mm),
  ) %>%
  arrange(desc(n_species))
order_summaries
```

```
## # A tibble: 22 × 5
##    Order    n_species median_max_longevity mean_max_longevity median_body_length
##    <chr>        <int>                <dbl>              <dbl>              <dbl>
##  1 Carnivo…        73                 19                 20.7               625.
##  2 Artioda…        52                 21.9               22.7              1362.
##  3 Rodentia        51                  7.3                9.45              159.
##  4 Primates        50                 33.2               33.8               495.
##  5 Chiropt…        11                 17                 20.0                74.0
##  6 Diproto…         7                 17.8               21.1               646.
##  7 Perisso…         7                 39.6               39.0              2587.
##  8 Soricom…         7                  3.2                3.1                81.9
##  9 Dasyuro…         6                  7.1                7.92              254.
## 10 Cingula…         4                 18.6               18.0               445.
## # ℹ 12 more rows
```

Written note

We chose PanTheria and Anage, open source datasets examining different morphological and life history metrics across thousands of species. I sourced the actual data from Github URLs, which are copies of the original PanTheria (Jones et al. 2009) and AnAge (Human Ageing Genomic Resources) databases.

The orders that are best represented here are Carnivora, Artiodactyla, Rodentia, and Primates. We are most interested in Carnivora, with a median max longevity of 19 years, and a median body length of 624 mm, and Primates with a median body length of 495 mm and median max longevity of 33 years.

in order to join the datasets in task 2, i had to make sure that the column names were the same so that they could be recognized across both datasets.

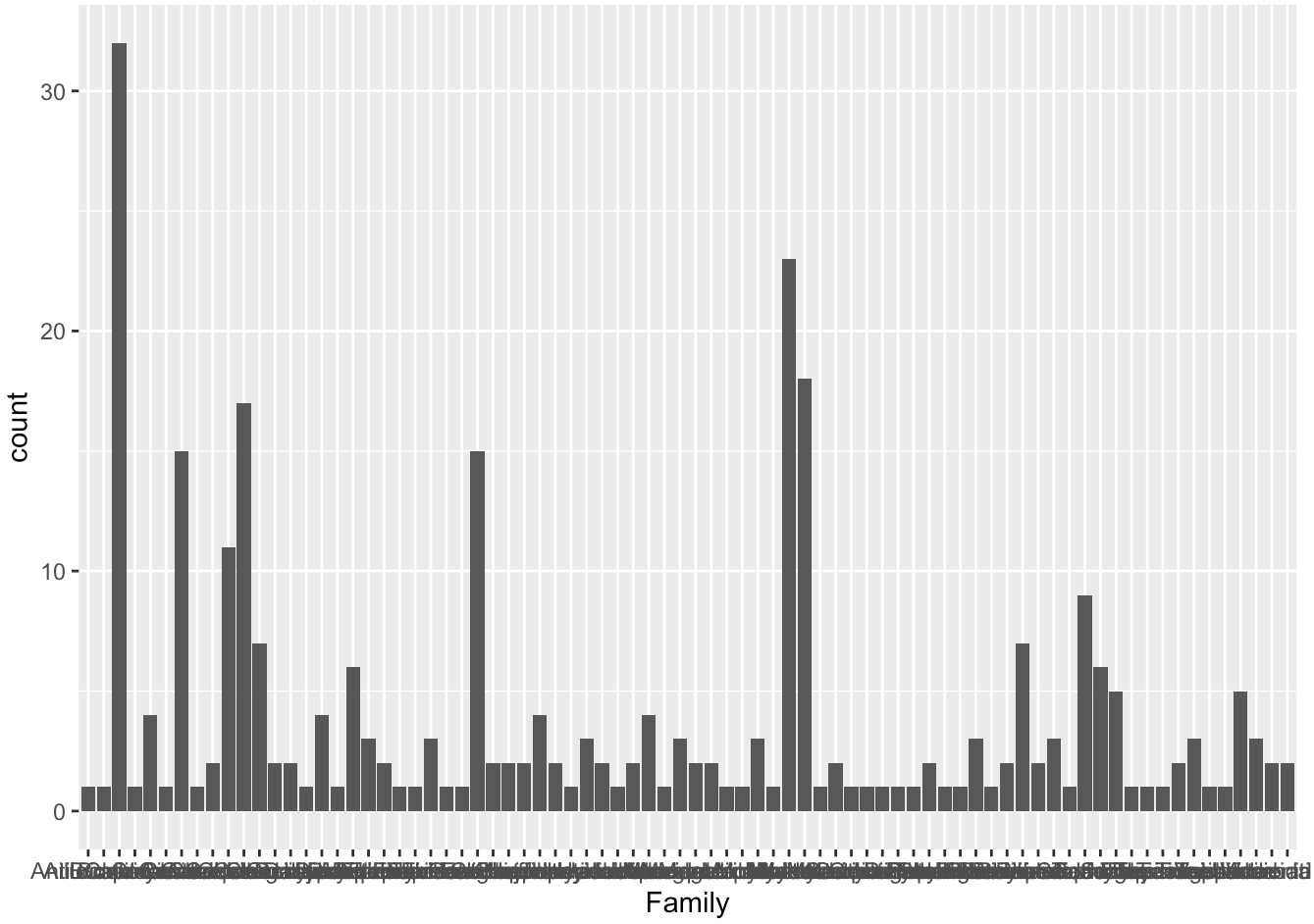
Part 2: Descriptive statistics (Week 4)

For the rest of the project, We use “pantheria_anage_joined.csv”, the joined dataset created in Part 1.

Task 1: Describe a categorical variable

```
pantheria_anage <- read.csv("pantheria_anage_joined.csv")
view(pantheria_anage)

pantheria_anage %>%
  ggplot() +
  geom_bar(
    mapping = aes(x= Family)
  )
```



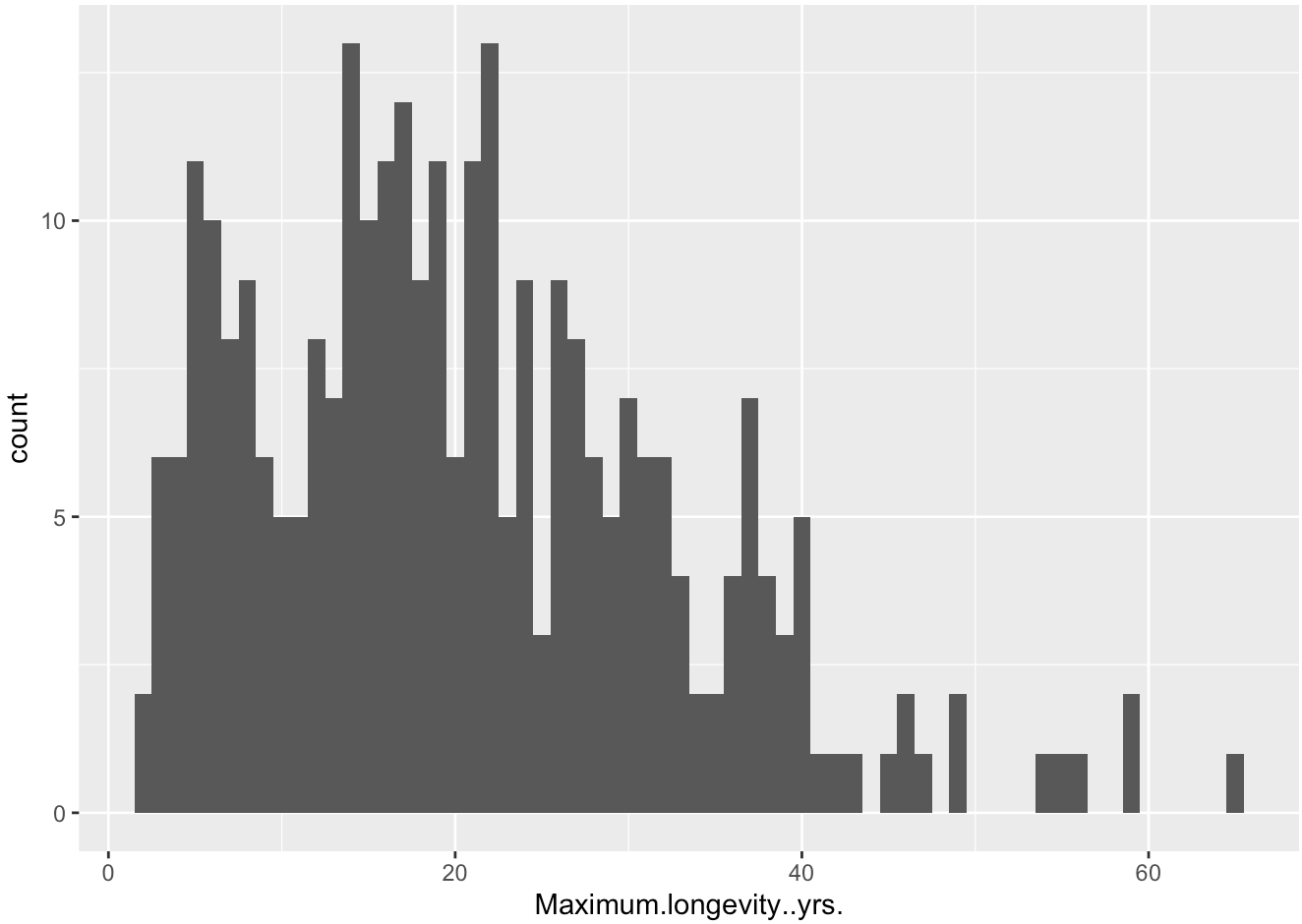

```
pantheria_anage %>%
  group_by(Family) %>%
  summarize(count = n())
```

```
## # A tibble: 78 × 2
##    Family         count
##    <chr>          <int>
##  1 Ailuridae          1
##  2 Antilocapridae     1
##  3 Bovidae           32
##  4 Burramyidae        1
##  5 Callitrichidae     4
##  6 Camelidae          1
##  7 Canidae           15
##  8 Castoridae         1
##  9 Caviidae           2
## 10 Cebidae           11
## # ℹ 68 more rows
```

In the dataset, “pantheria_anage_joined.csv”, the family “Bovidae” is most well-represented, with 32 entries (species) in this category. In contrast, many families are only represented by one entry.

Task 2: Describe a continuous variable

```
pantheria_anage %>%
  ggplot() +
  geom_histogram(
    mapping = aes(x= Maximum.longevity..yrs.),
    binwidth = 1  )
```



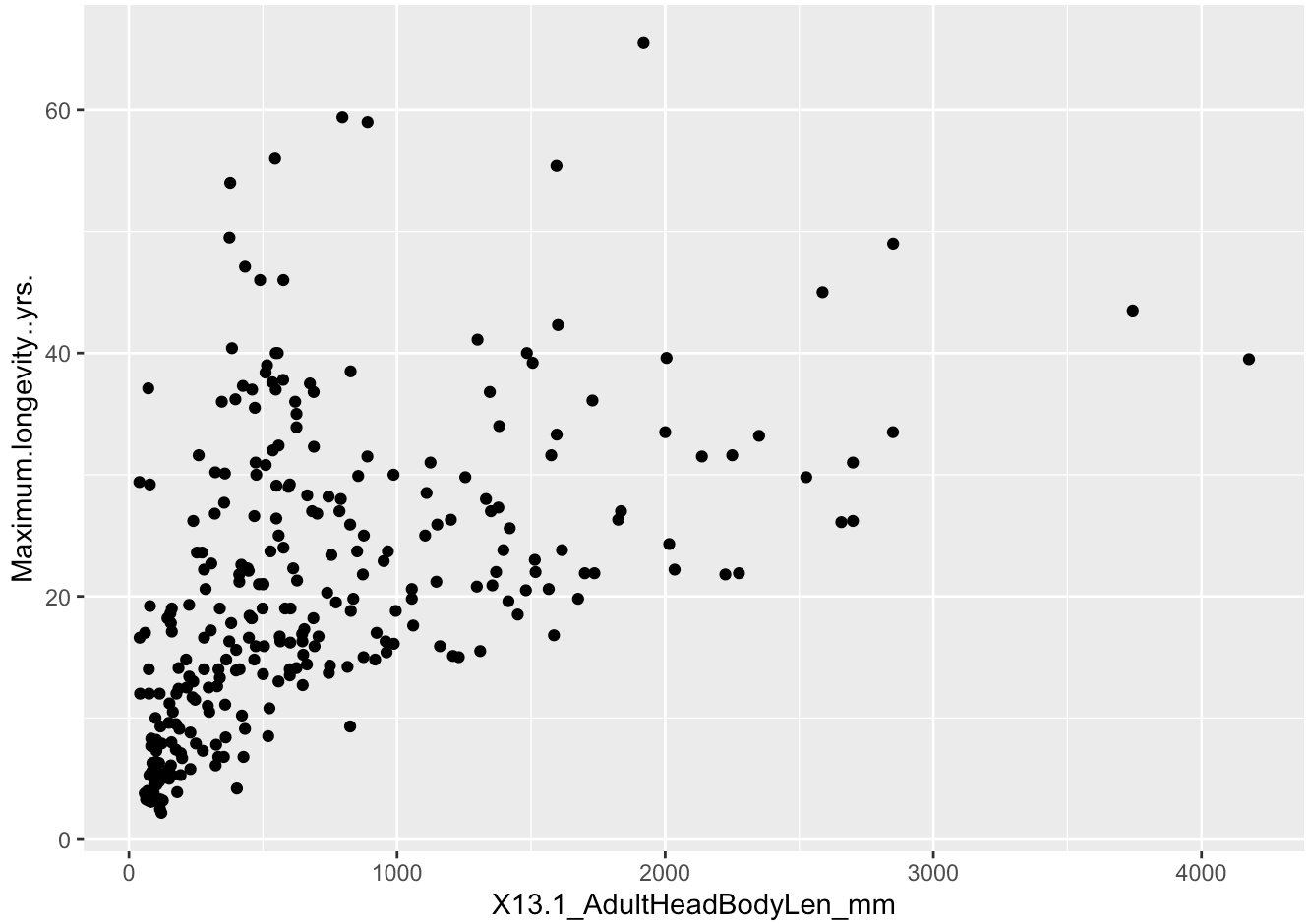

```
summary(pantheria_anage$Maximum.longevity..yrs.)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    2.20   12.00   19.00   20.85   28.20   65.50
```

In the dataset, “pantheria_anage_joined.csv”, maximum longevity is not normally distributed, and rather, has a rightward skew. This means that a five number summary is more appropriate here than mean and standard deviation. Across the dataset, the min maximum longevity is 2.2 years, the median maximum longevity is 19 years, and the max maximum longevity is 65.5 years.

Task 3: Relationship between two variables

```
pantheria_anage %>%
  ggplot() +
  geom_point(
    mapping = aes(x = X13.1_AdultHeadBodyLen_mm, y = Maximum.longevity..yrs.)
  )
```



In the dataset, “pantheria_anage_joined.csv”, adult body length including the head is positively correlated with maximum longevity of that species. However, several species exhibit above median maximum longevity (>19 years) despite having a low adult body length. This suggests that in our dataset, body length alone does not account for differences in maximum longevity between species. This may be better understood by looking at additional variables in the dataset, such as body mass, average latitude, etc.

Part 3: Inferential statistics (Week 5)

Task 1: State your hypotheses

To test whether evolutionary history is associated with longevity, I compare two orders. I chose Primates and Carnivora because they are two of the most well-represented orders in the dataset (see the order summary table in Part 1), which gives each group enough species to compare.

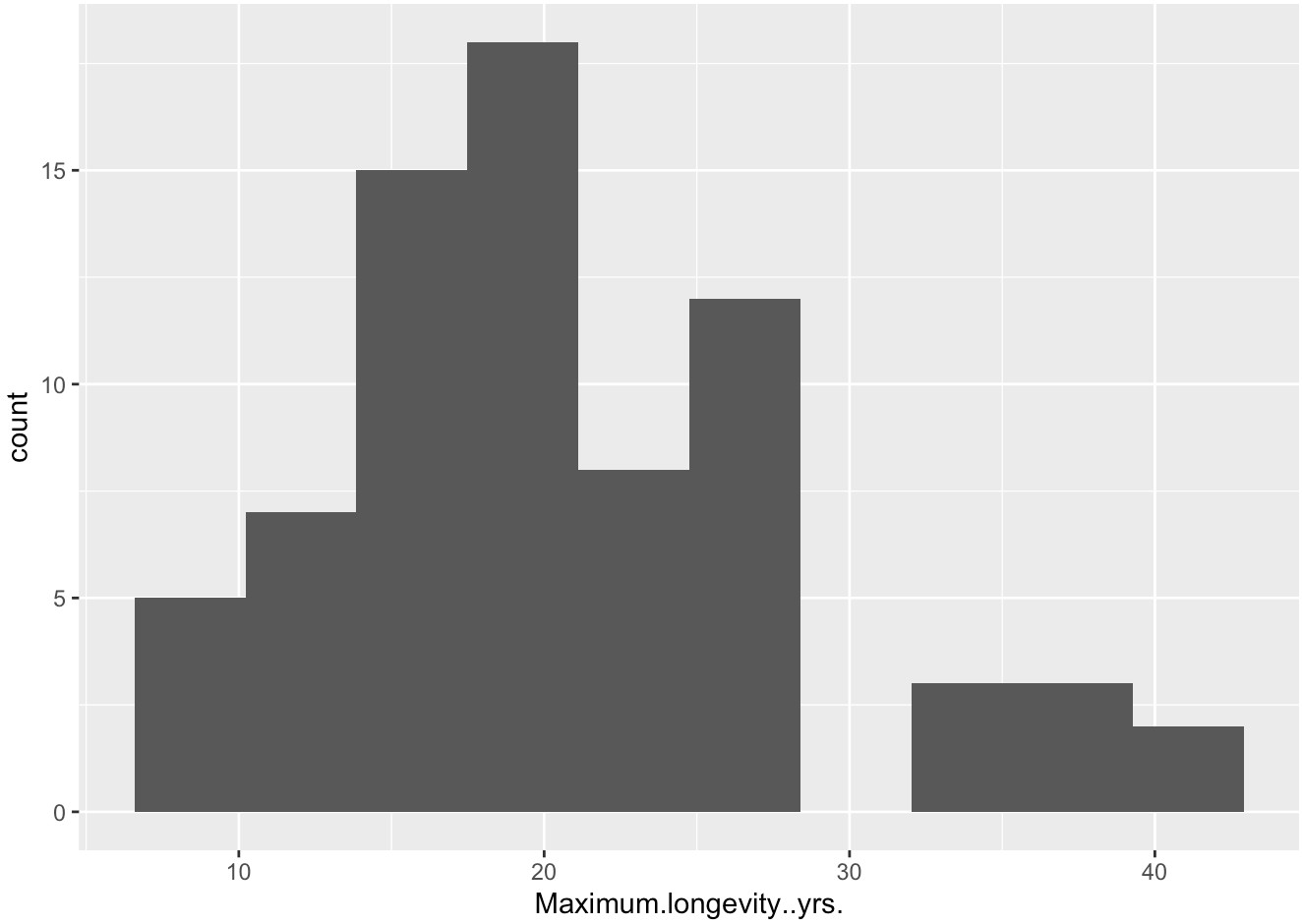
H_0: Primates and Carnivora have the same distribution of max lifespan

H_a: Primates and Carnivora have different distributions of max lifespan

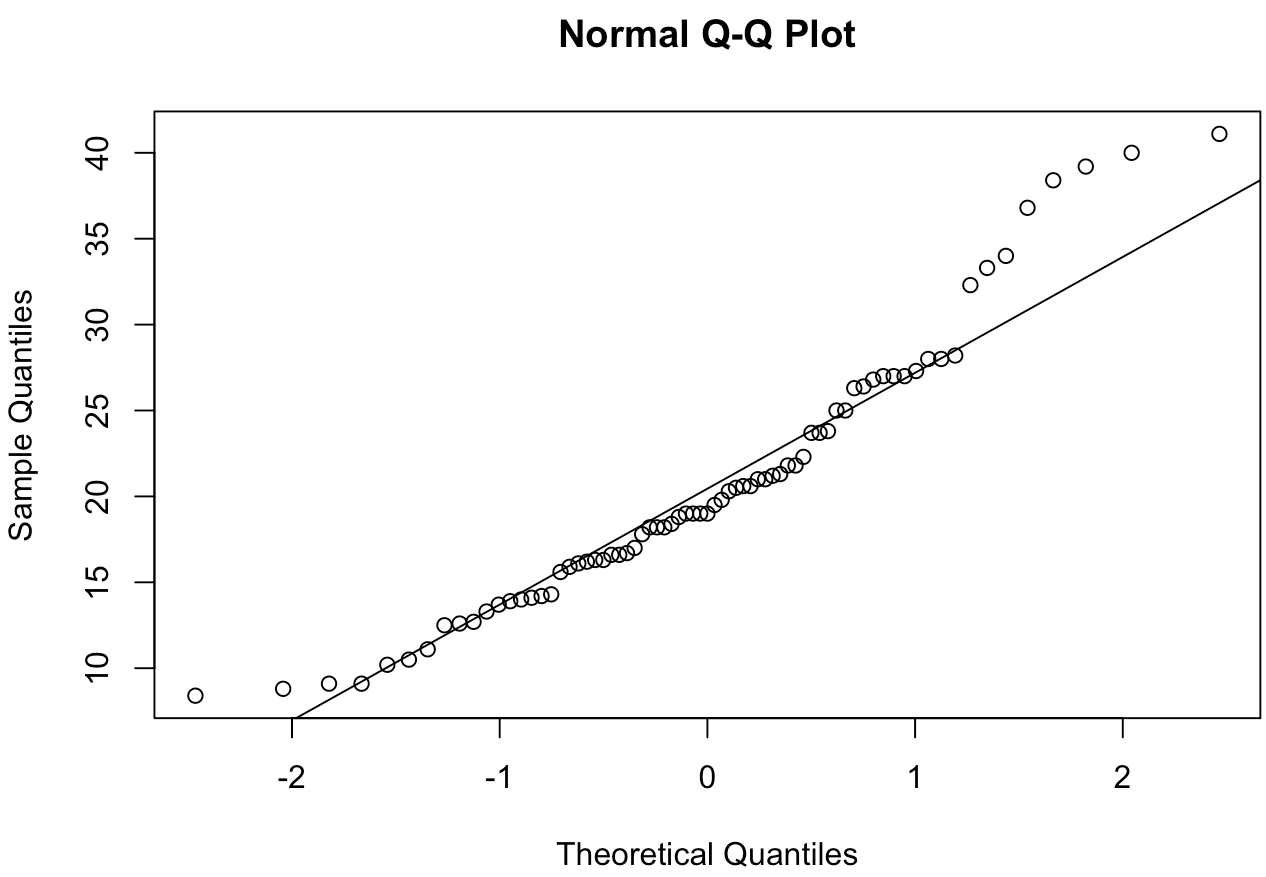
Task 2: Check normality

```
library(tidyverse)
library(ggplot2)
agingpantheria <- read.csv("pantheria_anage_joined.csv")
view(agingpantheria)
```

```
##Carnivora
agingpantheria %>%
  filter(Order == "Carnivora") %>%
  ggplot() +
    geom_histogram(aes(x = Maximum.longevity..yrs.), bins = 10)
```



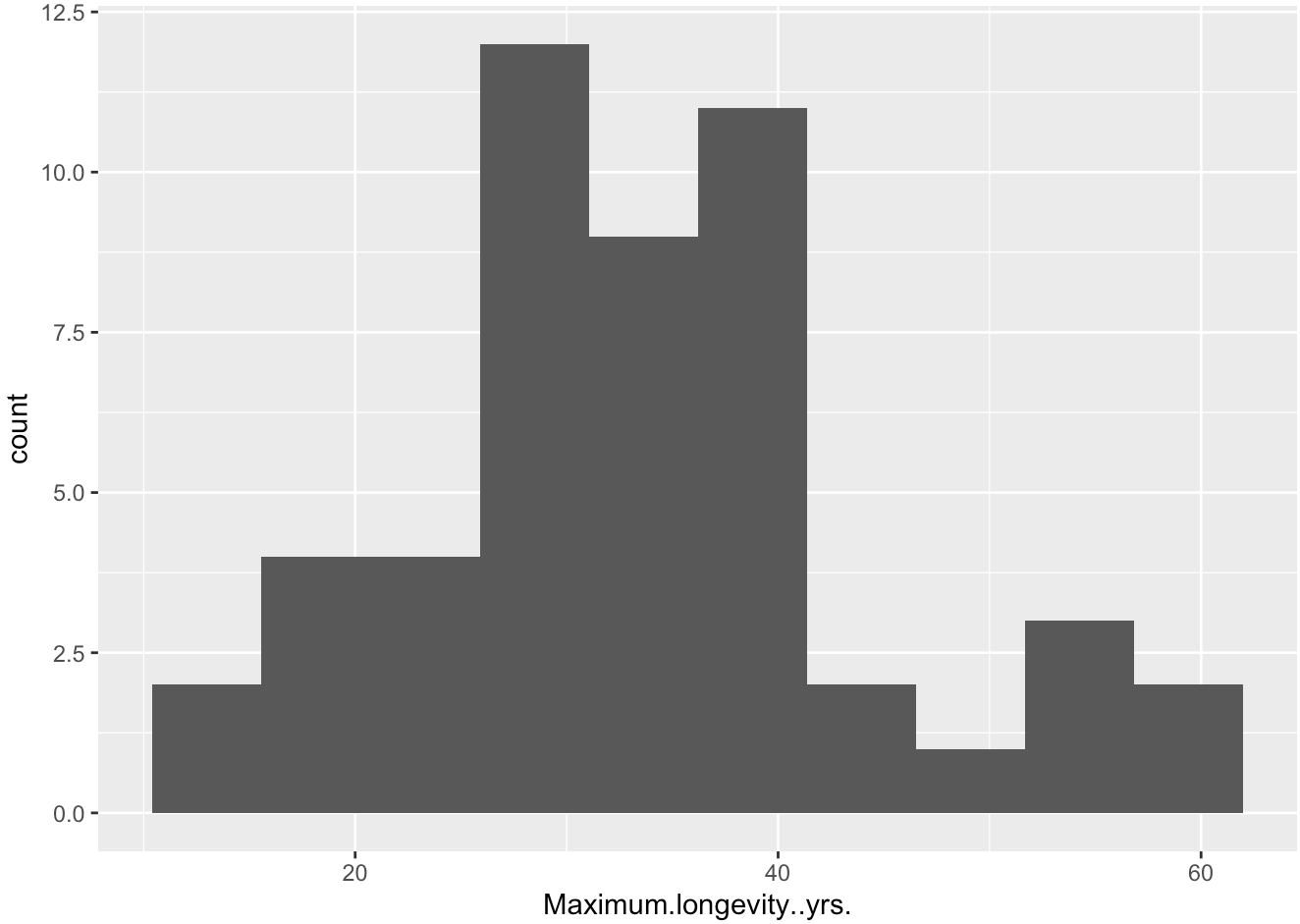

```
agingpantheria %>%
  filter(Order == "Carnivora") %>%
pull(Maximum.longevity..yrs.) %>%
  qqnorm()
qqline(agingpantheria %>% filter(Order == "Carnivora") %>% pull(Maximum.longevity..yr
s.))
```



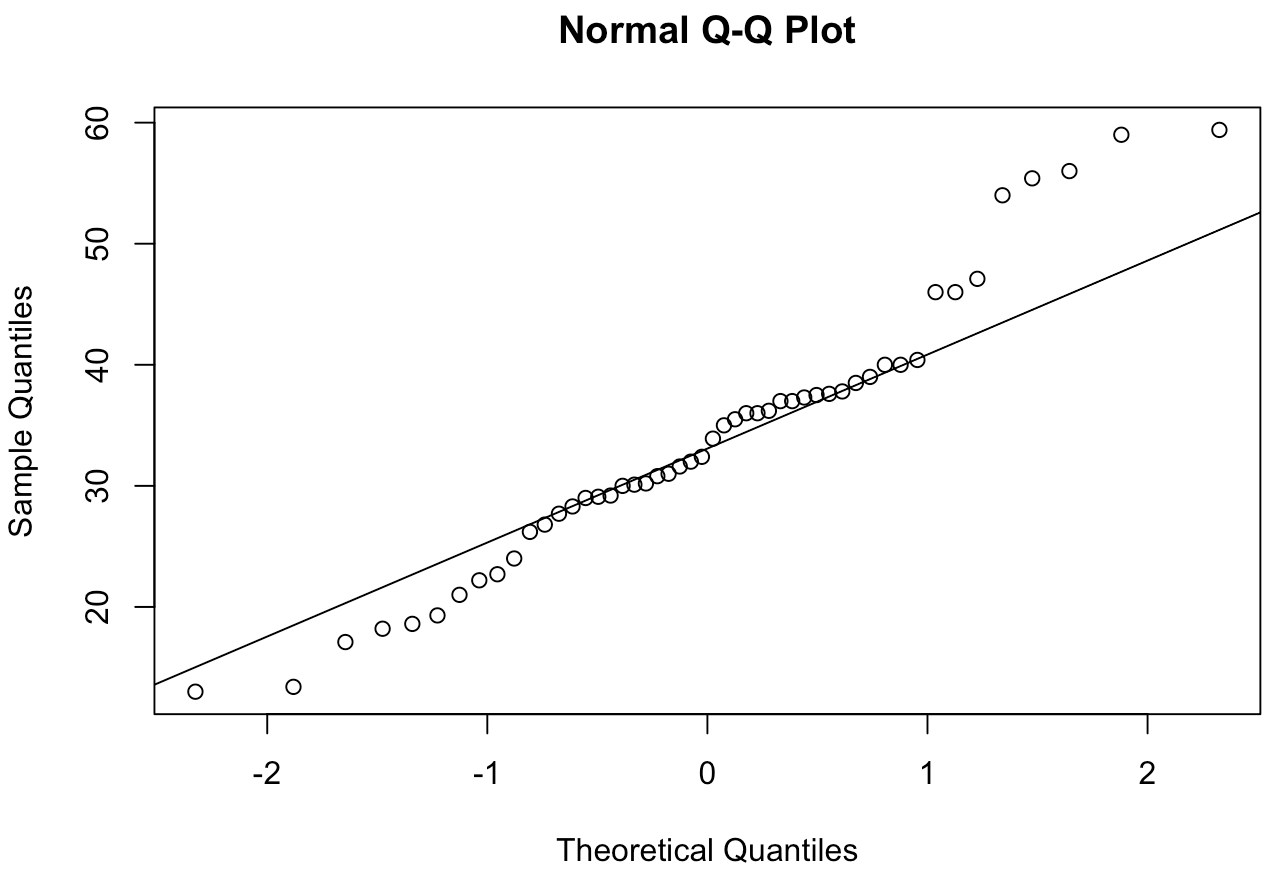

```
carnivore_longevity <- agingpantheria$Maximum.longevity..yrs.[agingpantheria$Order == "C
arnivora"]
shapiro.test(carnivore_longevity)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  carnivore_longevity
## W = 0.94349, p-value = 0.002623
```

```
##Primates
agingpantheria %>%
  filter(Order == "Primates") %>%
  ggplot() +
    geom_histogram(aes(x = Maximum.longevity..yrs.), bins = 10)
```




```
agingpantheria %>%
  filter(Order == "Primates") %>%
pull(Maximum.longevity..yrs.) %>%
  qqnorm()
qqline(agingpantheria %>% filter(Order == "Primates") %>% pull(Maximum.longevity..yrs.))
```




```
Primate_longevity <- agingpantheria$Maximum.longevity..yrs.[agingpantheria$Order == "Pri
mates"]
shapiro.test(Primate_longevity)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  Primate_longevity
## W = 0.9623, p-value = 0.1109
```

the shapiro-wilk test reveals that while primate longevity does not significantly deviate from normality (p = 0.11), carnivore longevity does (p = 0.0026). Therefore, I cannot confidently use a statistical test that assumes normality. Therefore, I will use the Wilcoxon rank-sum test.

Task 3: Run the appropriate test

```
CarnivoresvPrimates <- agingpantheria %>%
  filter(Order %in% c("Carnivora", "Primates"))

wilcox.test(Maximum.longevity..yrs. ~ Order, data = CarnivoresvPrimates, exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  Maximum.longevity..yrs. by Order
## W = 592, p-value = 2.199e-10
## alternative hypothesis: true location shift is not equal to 0
```

```
median(Primate_longevity)
```

```
## [1] 33.15
```

```
median(carnivore_longevity)
```

```
## [1] 19
```

To check whether primates are shorter than carnivores in this dataset, I compare the median adult body length of each order.

```
CarnivoresvPrimates %>%
  group_by(Order) %>%
  summarize(n_species = n(),
            median_body_length_mm = median(X13.1_AdultHeadBodyLen_mm))
```

```
## # A tibble: 2 × 3
##   Order     n_species median_body_length_mm
##   <chr>         <int>                 <dbl>
## 1 Carnivora        73                  625.
## 2 Primates         50                  495.
```

Task 4: Written conclusion

Since one of my groups (carnivores) did not have a normal distribution, I decided it would be more robust to use a Wilcoxon rank-sum test instead of a t-test, which assumes normality. I used the Wilcoxon rank-sum test to compare the max longevity of carnivores and primates, two orders within the class Mammalia. The p-value is 2.199e-10, which indicates a significant difference in longevity between carnivores and primates within this dataset; I reject the null hypothesis. In plain language, primates tend to live longer than carnivores (median max longevity: 33.2 vs. 19.0 years). This is an interesting finding because within the dataset, primates tend to be shorter than carnivores [median body length: 625 mm vs. 495 mm], but in the descriptive analyses in Part 2, there appeared to be a positive correlation between body length and max longevity; therefore, this between-order trend is the opposite of what I saw in Part 2. One potential explanation is that within orders, animals with longer body lengths tend to have higher max longevity, but that other factors drive variation between orders.

However, it is important to consider error. Since my p-value indicates that I can reject the null hypothesis, it is possible that I am making a Type I error, or a false positive.

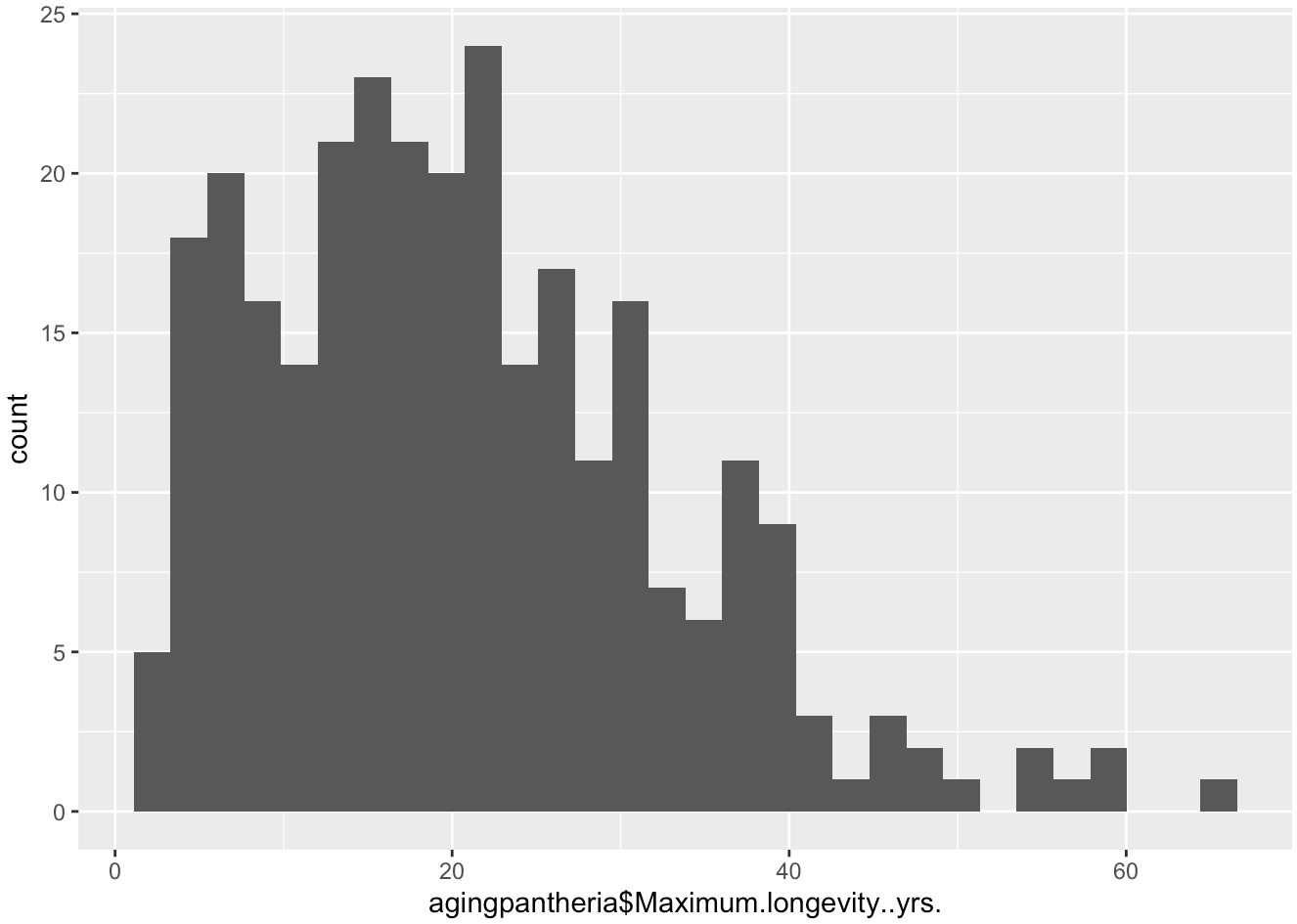
Part 4: Regression (Week 6)

```
library(tidyverse)
library(ggplot2)
agingpantheria <- read.csv("pantheria_anage_joined.csv")
view(agingpantheria)
```

Task 1: Identify your outcome variable

I have chosen to use maximum longevity as my outcome variable, which is a continuous variable. Since it is a continuous variable, I have chosen to use linear regression.

```
##MaxLongev
  ggplot() +
    geom_histogram(aes(x = agingpantheria$Maximum.longevity..yrs.), bins = 30)
```



Task 2: Fit the model

```
# choosing carnivora as reference order since it is the largest/most well-represented in the dataset
agingpantheria$Order <- relevel(as.factor(agingpantheria$Order), ref = "Carnivora")

# Predict max longevity from adult body length and order
model_multi <- lm(Maximum.longevity..yrs. ~ X13.1_AdultHeadBodyLen_mm + Order, data = agingpantheria)
summary(model_multi)
```

```
##
## Call:
## lm(formula = Maximum.longevity..yrs. ~ X13.1_AdultHeadBodyLen_mm +
##     Order, data = agingpantheria)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18.3486  -3.6557  -0.5002   2.8138  22.5142
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 13.439053   1.077712  12.470  < 2e-16 ***
## X13.1_AdultHeadBodyLen_mm    0.009957   0.001012   9.840  < 2e-16 ***
## OrderAfrosoricida           -2.115328   3.992582  -0.530 0.596682
## OrderArtiodactyla           -5.369058   1.428718  -3.758 0.000211 ***
## OrderChiroptera              5.728815   2.265585   2.529 0.012030 *
## OrderCingulata              -0.596851   3.451339  -0.173 0.862836
## OrderDasyuromorphia         -8.500458   2.881860  -2.950 0.003465 **
## OrderDidelphimorphia       -10.998792   3.983235  -2.761 0.006159 **
## OrderDiprotodontia           1.789280   2.657990   0.673 0.501424
## OrderErinaceomorpha         -4.100607   6.771884  -0.606 0.545340
## OrderHyracoidea             -4.055161   4.814059  -0.842 0.400344
## OrderLagomorpha             -8.877540   6.765910  -1.312 0.190618
## OrderMacroscelidea          -6.668394   4.839254  -1.378 0.169368
## OrderMonotremata            18.657034   4.819383   3.871 0.000136 ***
## OrderPeramelemorphia       -10.362211   4.823863  -2.148 0.032608 *
## OrderPerissodactyla         -0.747739   3.283882  -0.228 0.820055
## OrderPilosa                 11.431342   4.810964   2.376 0.018205 *
## OrderPrimates               15.519938   1.255158  12.365  < 2e-16 ***
## OrderProboscidea            32.951051   6.859962   4.803 2.61e-06 ***
## OrderRodentia               -6.362823   1.321126  -4.816 2.46e-06 ***
## OrderScandentia             -2.881139   6.775974  -0.425 0.671035
## OrderSoricomorpha          -11.270523   2.730993  -4.127 4.92e-05 ***
## OrderTubulidentata           3.870312   6.774371   0.571 0.568267
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.708 on 266 degrees of freedom
## Multiple R-squared:  0.7155, Adjusted R-squared:  0.692
## F-statistic: 30.41 on 22 and 266 DF,  p-value: < 2.2e-16
```

Task 3: Interpret your regression coefficients

predictor coefficient 1: X13.1_AdultHeadBodyLen_mm

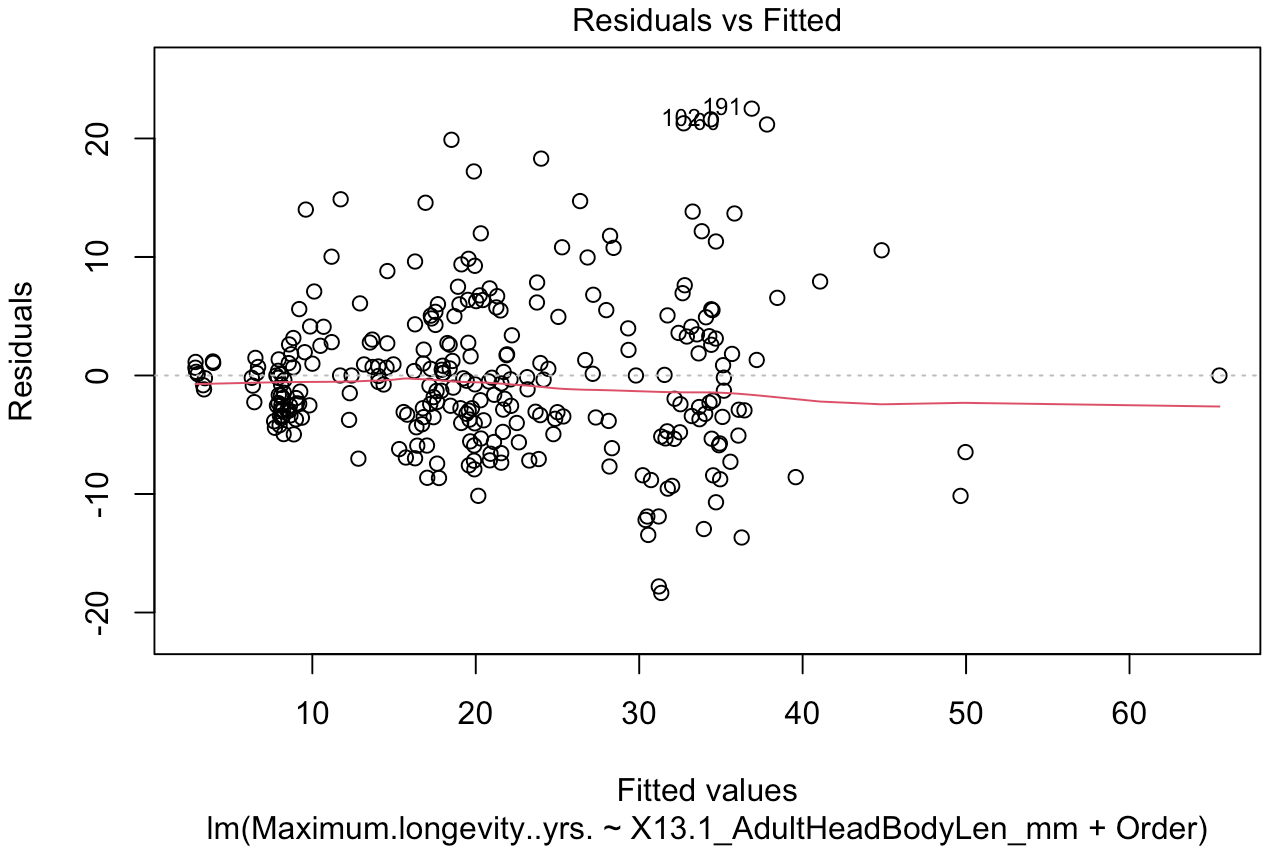
Adult body length in mm predicting maximum longevity in years. Holding taxonomic Order constant, for every 1 mm increase in adult body length, maximum longevity increases by 0.00996 years (~4 days) on average. This predictor is statistically significant (p < 2e-16)

predictor coefficient 2: OrderPrimates

Taxonomic Order (primates) predicting maximum longevity in years. Holding adult body length constant, primates are expected to live 15.5 years longer on average compared to species in the reference Order, Carnivora. This predictor is statistically significant ($p < 2e\text{-}16$).

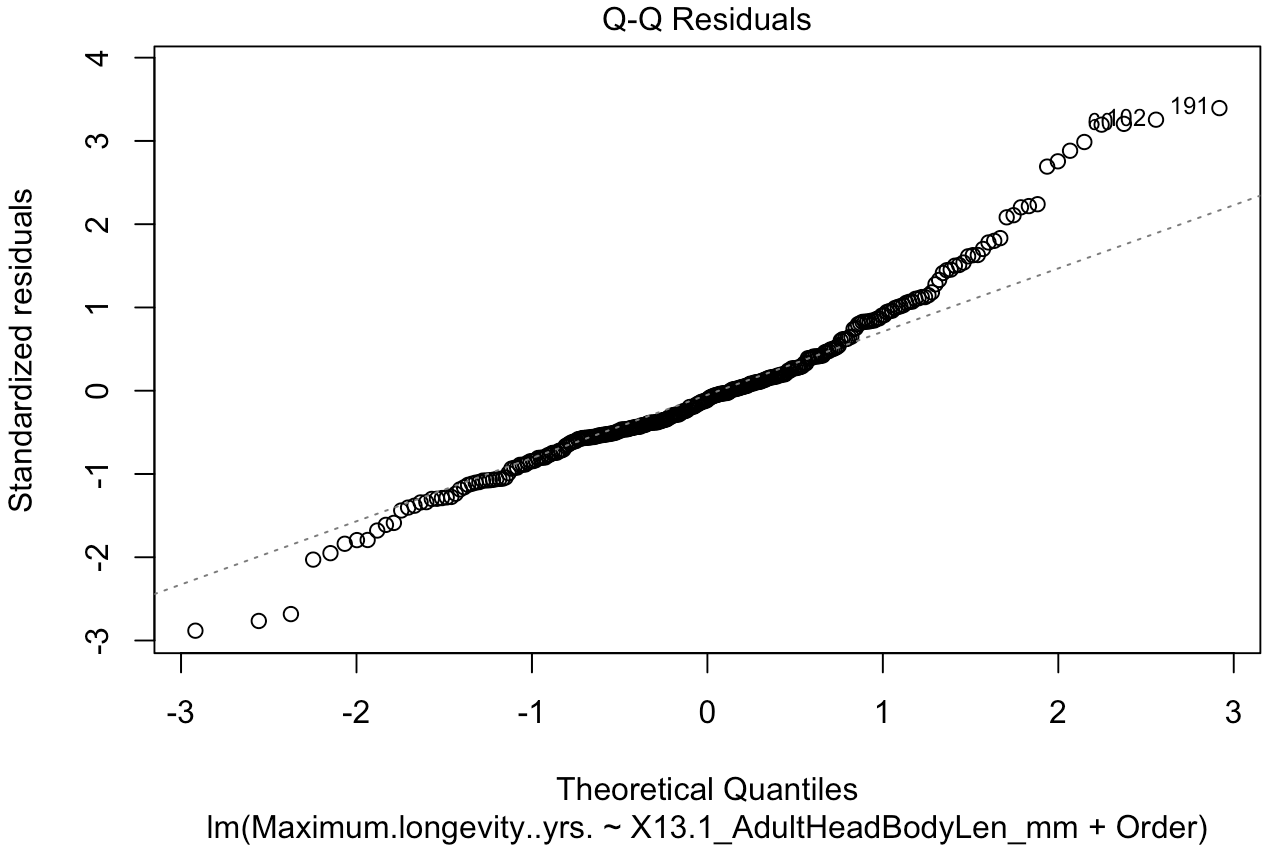
Task 4: Run one diagnostic check

```
# Residual plots
plot(model_multi, which = 1)   # Residuals vs. Fitted
```




```
plot(model_multi, which = 2)   # Normal QQ plot
```

```
## Warning: not plotting observations with leverage one:
##   25, 39, 184, 249, 274
```



residuals vs fitted plot: There is a funnel shape from left to right, indicating uneven spread. Lower fitted values have less vertical spread and as fitted values increase there is more spread. This indicates non-constant variance, violating the constant variance assumption of linear regression. However, since the red line stays relatively flat and close to zero, this suggests a reasonably centered general linear trend.

normal dist in the Q-Q plot: While the central residuals follow the reference line closely, the upper tail curves upward substantially, indicating a heavy right-skew in the residuals and suggesting that the residuals are not normally distributed.

Written paragraph

I fitted a multiple linear regression model to predict maximum longevity from adult body length and taxonomic order. Key results indicate that adult body length is a statistically significant positive predictor of longevity, with max longevity increasing by about 4 days for every mm increase in body length on average according to the model. Additionally, the taxonomic order Primates exhibits a statistically significant positive effect, living 15.5 years longer on average than the baseline reference order, Carnivora, holding body length constant. Overall, the model explains about 72% of the variation in max longevity in this dataset ($R^2 = 0.72$, adjusted $R^2 = 0.69$).

however, diagnostic checks indicate notable concerns with the model assumptions. the residuals versus fitted plot shows a funnel-like pattern, indicating heteroscedasticity (increased error at higher predicted values). Also, the normal Q-Q plot shows an upper curved tail, indicating right-skewness and non-normally distributed residuals caused by outlier species with exceptionally long lifespans. These diagnostics suggest that applying a logarithmic transformation to the variables may stabilize variance and help this data meet linear model assumptions.

Another limitation is that five orders (Erinaceomorpha, Lagomorpha, Proboscidea, Scandentia, and Tubulidentata) are only represented by one species in the dataset. R flags these species as having “leverage one” in the diagnostic plots, meaning the model fits them exactly, so the coefficients for these orders are not meaningful. For example, the model estimates that Proboscidea live 33 years longer than Carnivora, but this is based on a single species.

Conclusion

Going back to my larger question of what drives longevity across clades, both evolutionary history and morphology appear to matter in this dataset. Adult body length is a statistically significant positive predictor of max longevity, but taxonomic order explains large differences beyond body length: holding body length constant, primates live about 15.5 years longer than carnivores on average, and together, body length and order explain about 72% of the variation in max longevity. However, taxonomic order is only a rough measure of evolutionary history, several orders are represented by only one species, and the model does not fully meet the constant variance and normality assumptions of linear regression, so these results should be interpreted with caution.